Guest Check-In System — Implementation Workflow

Technical documentation describing the state machine, agent execution pipeline, LLM fallback chain, and full onboarding flow.

Table of Contents

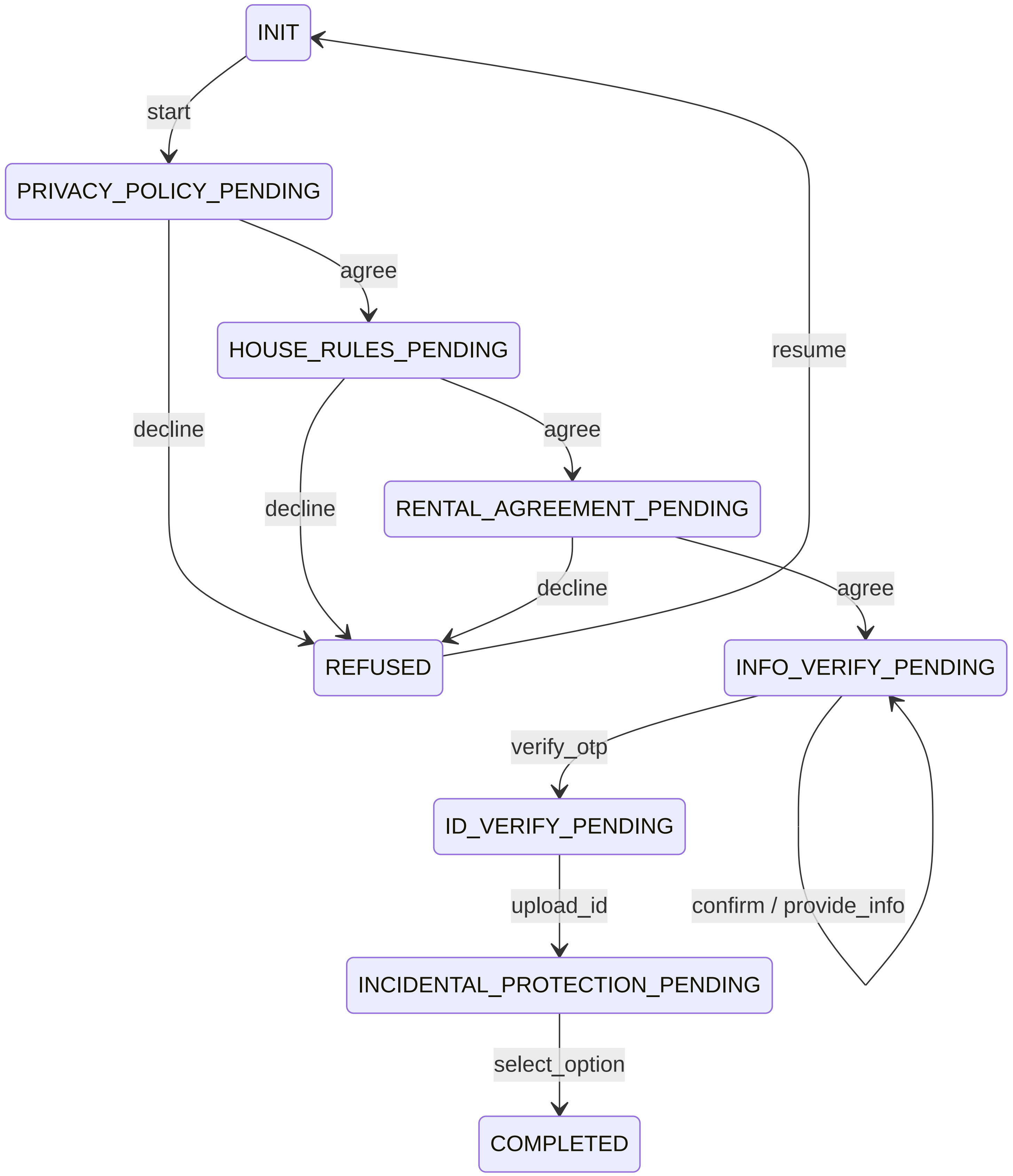
1. State Machine Diagram

The guest check-in workflow is governed by a 9-state finite state machine. Each state represents a required onboarding step; transitions are driven by **intents** detected from the guest's messages. The transition table is defined in `app/state_machine/transitions.py` and the state enum in `app/state_machine/states.py`.

States

State	Required Action	Valid Intents
`INIT`	Initialize check-in session	start
`PRIVACY_POLICY_PENDING`	Accept or decline Privacy Policy	agree, decline
`HOUSE_RULES_PENDING`	Accept or decline House Rules	agree, decline
`RENTAL_AGREEMENT_PENDING`	Accept or decline Rental Agreement	agree, decline
`INFO_VERIFY_PENDING`	Confirm personal info & verify OTP	confirm, provide_info, verify_otp
`ID_VERIFY_PENDING`	Upload government-issued ID	upload_id
`INCIDENTAL_PROTECTION_PENDING`	Select Damage Waiver or Security Hold	select_option
`COMPLETED`	None — onboarding finished	*(terminal)*
`REFUSED`	Guest declined an agreement	resume

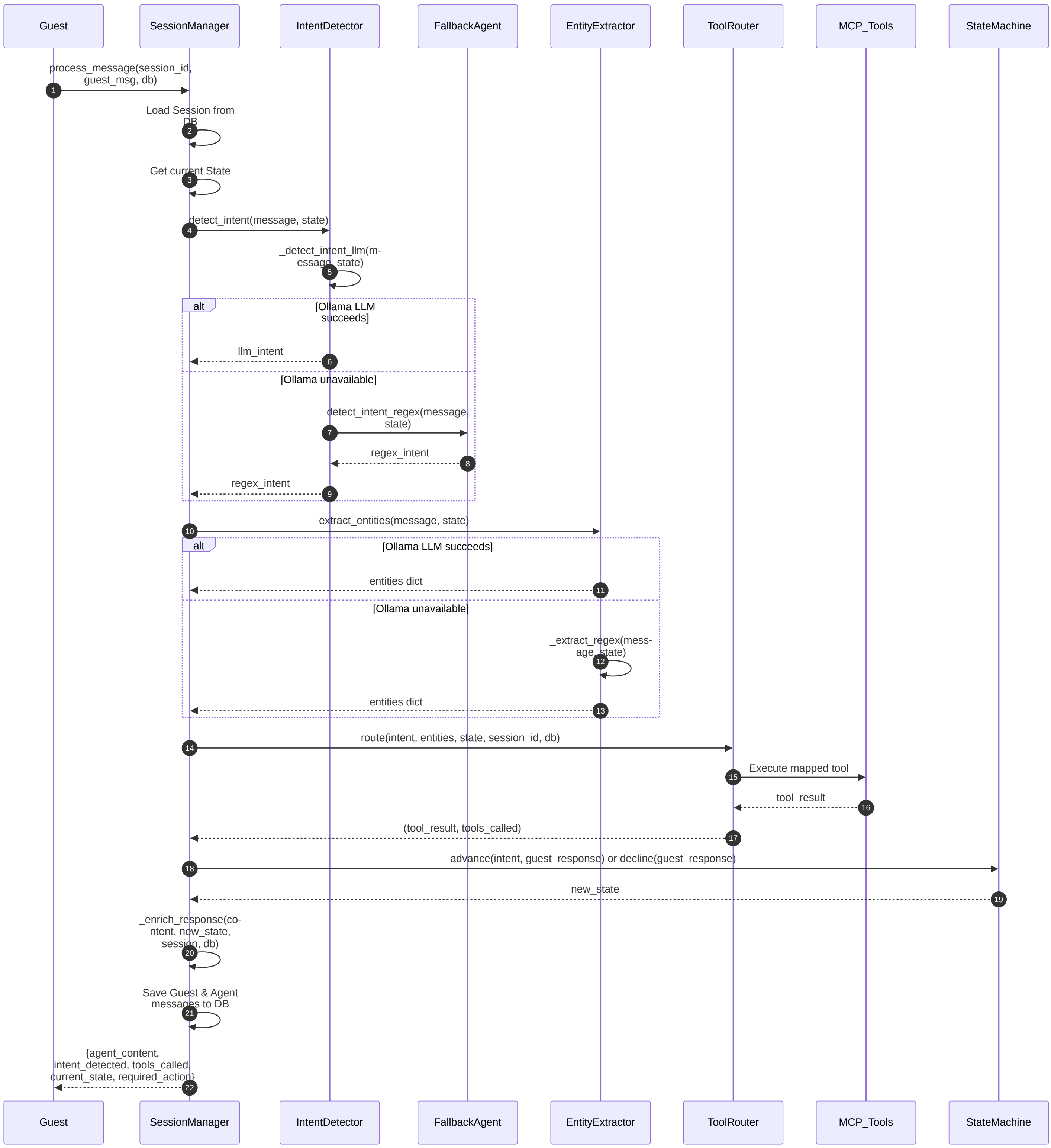
State Diagram



Note: The self-transitions on `INFO_VERIFY_PENDING` (`confirm` and `provide_info`) keep the guest in the same state. `confirm` triggers an OTP email; `provide_info` lets the guest correct their details before confirming. Only `verify_otp` (after successful OTP verification) advances to `ID_VERIFY_PENDING`.

2. Agent Execution Sequence

When a guest sends a message, the `SessionManager.process_message()` method orchestrates the full pipeline: load session → detect intent → extract entities → route to MCP tool → advance state → generate response. The sequence diagram below shows the interactions between the core components.

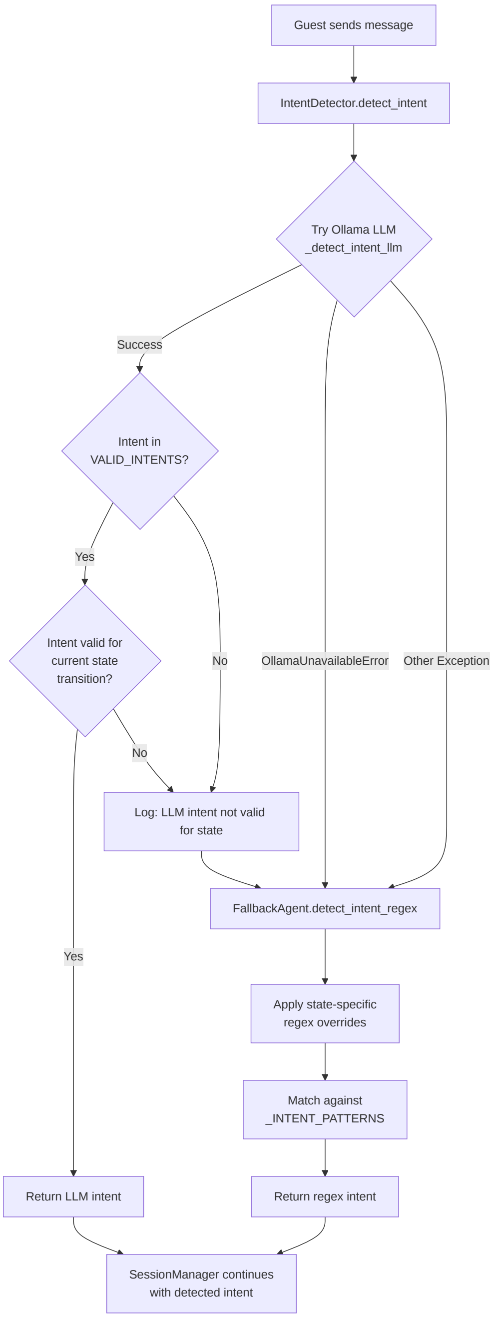


Key detail: The OTP verification gate lives inside `_advance_state()`. When the intent is `verify_otp` and the tool result indicates failure, the state machine does *not* advance — the guest stays in `INFO_VERIFY_PENDING` and receives an error message with remaining attempts.

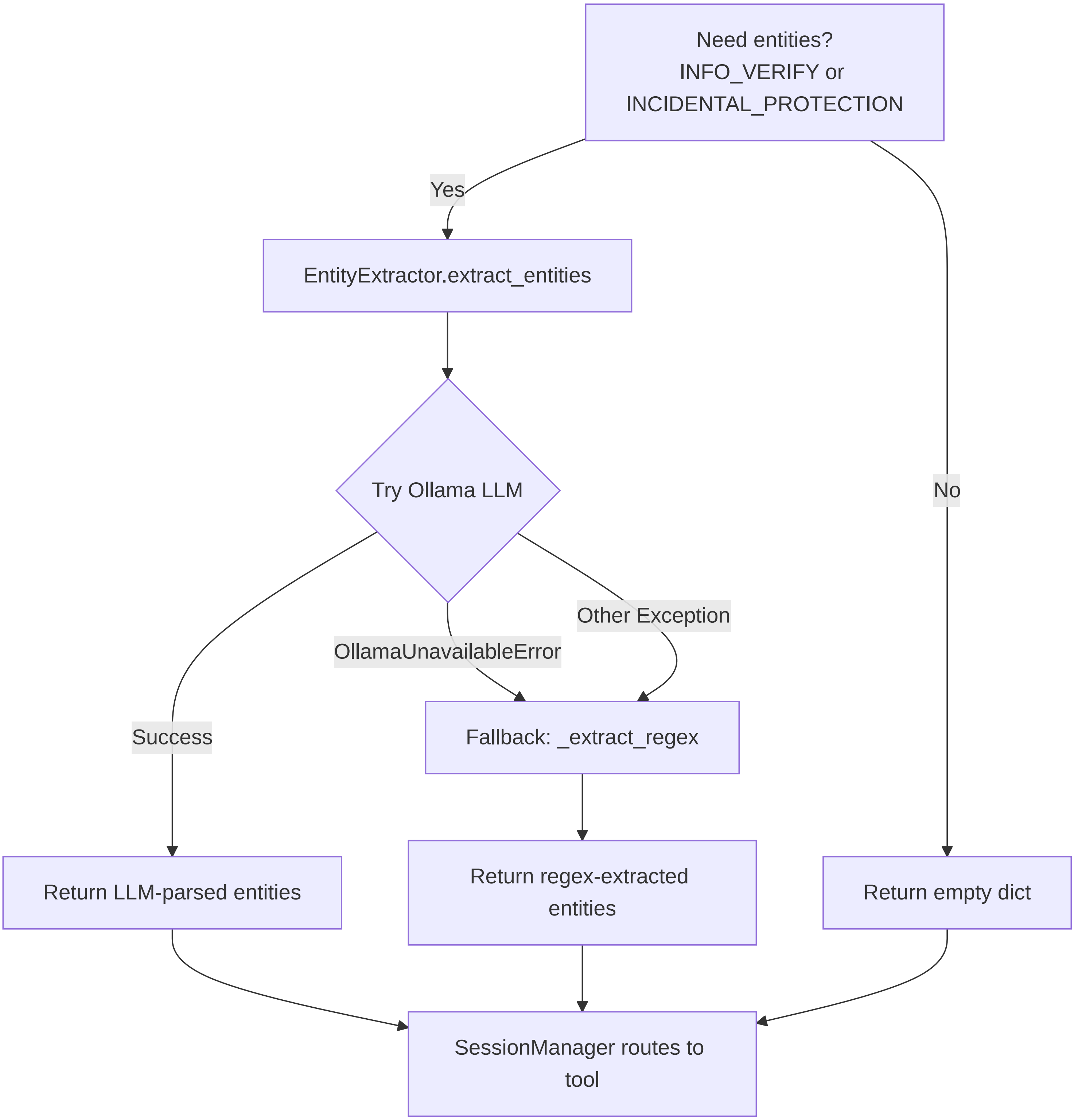
3. LLM Pipeline with Fallback

The system uses a primary Ollama LLM (Llama 3.1 8B) for intent detection and entity extraction. When Ollama is unavailable or returns an invalid intent, a regex-based `FallbackAgent` takes over. The flowchart below illustrates this two-tier strategy.

Intent Detection Pipeline



Entity Extraction Pipeline



FallbackAgent Regex Patterns

The `FallbackAgent` uses an ordered list of compiled regex patterns. The order matters — more specific patterns are checked first:

decline — "no", "declined", "refuse", "don't agree", etc.

request_help — "I need help", "speak to someone", etc.

select_option — "damage waiver", "security hold", "option 1/2"

provide_info — "my name is", "call me", "update", "change"

question — "what", "when", "how", or contains "?""

agree — "yes", "agree", "accept", "okay", "confirmed"

greeting — "hi", "hello", "hey", "good morning"

State-specific overrides run *before* the general patterns:

`INIT` → "start/begin/ready" → **start**

`INFO_VERIFY_PENDING` → "confirm/correct" → **confirm**; 6-digit number → **verify_otp**

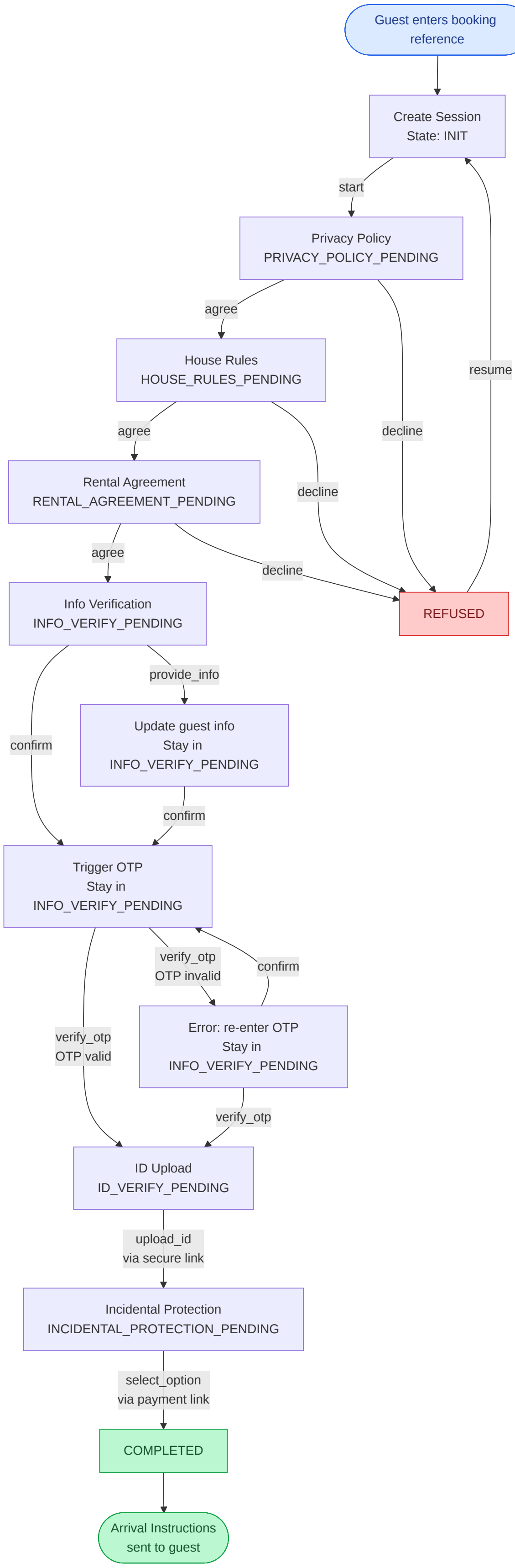
`ID_VERIFY_PENDING` → "uploaded/sent id" → **upload_id**

`INCIDENTAL_PROTECTION_PENDING` → "damage waiver/security hold" → **select_option**

`REFUSED` → "resume/restart" → **resume**

4. Full User Onboarding Flow

This flowchart shows the complete guest onboarding journey from start to finish, including all agreement steps, the OTP verification sub-flow, external links for ID upload and incidental protection, and the decline/resume cycle.



Step-by-Step Walkthrough

Session Creation — The guest provides a booking reference. A new session is created in the `INIT` state.

Privacy Policy — The agent presents the privacy policy text. The guest must `agree` or `decline`. Declining sends the guest to `REFUSED`.

House Rules — Same agree/decline pattern. Agreement is recorded via the `record_agreement` MCP tool.

Rental Agreement — Same agree/decline pattern. Third and final agreement step.

Info Verification — The agent retrieves the guest's reservation data (name, email, phone, guest count). The guest can:

`provide_info` — correct details (self-transition)

`confirm` — triggers OTP email (self-transition, stays in `INFO_VERIFY_PENDING`)

`verify_otp` — if OTP is valid, advances to `ID_VERIFY_PENDING`; if invalid, error with remaining attempts

ID Upload — The agent generates a secure, HMAC-signed, time-limited link for uploading a government-issued ID. The guest uploads externally and confirms with `upload_id`.

Incidental Protection — The agent generates a selection + payment link. The guest chooses between Damage Waiver ($49) or Security Hold ($250). Confirming with `select_option` completes onboarding.

Completed — Arrival instructions (property address, WiFi, lockbox code, check-in/out times) are fetched and delivered to the guest.

Resume from Refusal — At any `REFUSED` state, the guest can say "resume" to restart from `INIT`.

> **External flows:** ID upload and incidental protection both use HMAC-signed, time-limited links generated by `app/services/link_service.py`. The guest completes these steps outside the chat widget (in a browser), then returns to the chat to confirm completion.
